Technical Report: TDE-YOLOX v2.0

A Simpler Three-Component Reference Architecture for Adverse-Condition Zero-Shot Detection

Research implementation report

August 2026

Abstract

This report documents the simpler TDE-YOLOX v2.0 implementation built on YOLOX-S, with emphasis on exact code-level behavior, optimization pathways, reported epoch-30 results, and limitations. The architecture contains three experimental mechanisms: (i) a shallow functional test-time training stage in Dark2 that adapts GroupNorm-related parameters by one masked-denoising reconstruction step; (ii) a TRIBRID PAFPN in which each CSP aggregation block retains a dense local path and receives a low-amplitude sparse-global residual path; and (iii) an ENGRAM associative-memory augmentation applied only to the classification branch and weakly supervised by the same SimOTA foreground targets used by the detector. The reported ACDC condition-specific results show positive AP deltas on snow (+1.1 points), night (+1.3), and rain (+1.2), but a substantial fog regression (-5.1). Consequently, the architecture should be interpreted as a useful positive reference and source of robust design principles rather than as a universally superior adverse-weather detector. The report also identifies several implementation caveats that materially affect interpretation, including a mask-ratio naming discrepancy, an active TTT schedule that differs from its declared fields, hard Top-K routing whose score values are discarded, a mismatch between Engram retrieval and auxiliary-logit geometry, and a potential batch-order mismatch in memory-anchor supervision.

Contents

1 Purpose and Scope

The purpose of this report is to make the historical simpler TDE-YOLOX version reproducible and technically inspectable. It is not a description of the later, more complex TDE redesign. Instead, it records the architectural state that produced the supplied epoch-30 results and explains why this version is useful as a reference point for future work.

Three distinctions are maintained throughout:

1. **Implementation versus comments.** Where comments and executable behavior disagree, the report follows the executable behavior.
2. **Observation versus causality.** Aggregate and class-level AP changes are reported, but they are not attributed to a specific component without an ablation.
3. **Historical reproduction versus strict ZSDA.** The historical experiment configuration uses an adverse annotation file for validation/testing. Reproducing that configuration is distinct from a strict clean-source-only zero-shot protocol.

2 Problem Setting

Let the source-domain training set be

$$\mathcal{D}_s = \{(x_i, y_i)\}_{i=1}^{N_s}, \tag{1}$$

where x_i is a source image and y_i contains object classes and bounding boxes. The target domain consists of adverse-condition images

$$\mathcal{D}_t = \{x_j^t\}_{j=1}^{N_t}, \tag{2}$$

with snow, night, rain, and fog conditions used for evaluation.

The intended zero-shot domain adaptation objective is to learn on the source detector while improving target robustness without target-domain gradient supervision. In the historical implementation, three mechanisms are introduced around a YOLOX-S chassis:

$$f_{\text{TDE}} = f_{\text{head+Engram}} \circ f_{\text{Tribrid PAFPN}} \circ f_{\text{backbone+TTT}}. \tag{3}$$

The three components are intended to address different aspects of distribution shift:

- shallow feature adaptation through TTT;
- contextual information aggregation through sparse global attention;
- semantic identity restoration through associative memory.

3 Overall Architecture

The model uses YOLOX-S scale, with depth multiplier 0.33 and width multiplier 0.50. The backbone outputs Dark3, Dark4, and Dark5 features to a modified PAFPN. The head remains decoupled into regression/objectness and classification pathways, but the classification pathway receives Engram restoration.

Listing 1: High-level data flow

```
image
  -> Focus stem
  -> Dark2 [GN + optional functional TTT]
```

```
  -> Dark3 [standard BN]
  -> Dark4 [standard BN]
  -> Dark5 [standard BN]
  -> PAFPN with four C2f_Tribrid blocks
  -> P3/P4/P5
       -> regression + objectness [standard YOLOX]
       -> classification -> latent -> Engram -> restored classification
```

A defining characteristic of the simpler architecture is that experimental components are comparatively constrained. Dark2 alone is directly adapted. The sparse-global branch begins with a residual gate of approximately one percent. ENGRAM acts only on classification and receives an auxiliary loss weight of 0.05.

4 Component I: Functional Test-Time Training

4.1 Adaptive scope

The Focus stem remains standard. Dark2 is rebuilt from GroupNorm-aware convolution and CSP components, then wrapped inside `TTTAdaptiveStage`. Dark3–Dark5 use the standard BatchNorm-based path.

Let the Dark2 stage be

$$F_\phi(x) = \mathrm{Dark2}_\phi(x), \tag{4}$$

where only a subset

$$\phi_a \subset \phi \tag{5}$$

is adaptable. In code, parameters are selected if their names contain `gn` or `norm`. Each selected parameter receives a learned scalar inner-loop learning rate.

4.2 Projector

The auxiliary projector is:

$$P_\psi = \mathrm{Conv}_{1\times1} \to \mathrm{GN} \to \mathrm{GELU} \to \mathrm{GRN} \to \mathrm{Conv}_{1\times1}. \tag{6}$$

It also owns a learned mask token m.

4.3 Variance-based mask

For initial feature tensor

$$F_0 = F_\phi(x), \tag{7}$$

the implementation computes channel variance

$$V(h, w) = \mathrm{Var}_c\left(F_0(c, h, w)\right), \tag{8}$$

then applies 4×4 average pooling and obtains the 0.85 quantile $q_{0.85}$:

$$M(h, w) = \nVdash\left[\bar{V}(h, w) \geq q_{0.85}\right]. \tag{9}$$

Important implementation interpretation. Although comments call this “85% masking,” the inequality masks approximately the *top 15%* of pooled variance values under a continuous score distribution. Thus the historical TTT intervention is materially milder than the comments imply.

4.4 Inner objective

A Gaussian noise map

$$\epsilon \sim \mathcal{N}(0, \sigma^2), \qquad \sigma = 0.08 \tag{10}$$

is sampled in the experiment configuration. The current feature is corrupted:

$$\tilde{F} = (F_\phi(x) + \epsilon) \odot (1 - M) + m \odot M. \tag{11}$$

The projector reconstructs the original current-image feature:

$$\hat{F} = P_\psi(\tilde{F}), \tag{12}$$

and the inner loss is

$$\mathcal{L}_{\mathrm{TTT}} = \frac{1}{|\Omega|} \sum_{u \in \Omega} \left\| \hat{F}_u - \mathrm{sg}(F_{0,u}) \right\|_2^2. \tag{13}$$

The one-step update is

$$\phi_a' = \phi_a - \alpha_{\phi_a} \nabla_{\phi_a} \mathcal{L}_{\mathrm{TTT}}, \tag{14}$$

where α_{ϕ_a} is learned.

The final output for that image is computed by a functional call with ϕ_a'. Base weights are not overwritten, so adaptation is episodic.

4.5 Inference behavior

At inference, the wrapper returns TTT probability 1.0. Unless explicitly bypassed, every image therefore triggers one functional adaptation step.

4.6 Training-time probability schedule

The wrapper declares:

$$p_{\mathrm{start}} = 0.1, \quad p_{\mathrm{end}} = 0.5, \quad e_{\mathrm{warmup}} = 10, \quad e_{\mathrm{ramp}} = 60. \tag{15}$$

However, the active implementation after warm-up is

$$p(e) = 0.1 + (0.7 - 0.1) \frac{e - 10}{E_{\max} - 10}, \tag{16}$$

so `ramp_end` and `prob_end` are not used by the active formula. The trainer additionally updates the epoch state twice, which can introduce an off-by-one schedule discrepancy.

5 Component II: Tribrid Sparse-Global PAFPN

5.1 C2f_Tribrid topology

Each of the four PAFPN aggregation CSP blocks is replaced by `C2f_Tribrid`. The block preserves a dense local path and adds a global path:

$$G(x) = \mathrm{DSA} \circ \mathrm{MBConv} \circ \mathrm{CoordAttn} \circ \mathrm{SimAM}(x). \tag{17}$$

The final local feature is updated by

$$y_{\mathrm{out}} = y_{\mathrm{local}} + \tanh(g)\, G(y_{\mathrm{local}}), \tag{18}$$

with

$$g_0 = 0.01. \tag{19}$$

Since $\tanh(0.01) \approx 0.01$, the global branch starts at roughly one percent amplitude.

This is a strong fail-safe property: the local detector remains dominant until optimization increases the gate.

5.2 Sparse attention

For feature map $X \in \mathbb{R}^{B\times C\times H\times W}$,

$$N = HW, \qquad K = \max(1, \lfloor 0.1N \rfloor). \tag{20}$$

An indexer produces spatial scores

$$s = I(X) \in \mathbb{R}^{B\times N}. \tag{21}$$

Hard Top-K indices are selected:

$$\mathcal{I}_K = \mathrm{TopKIndices}(s, K). \tag{22}$$

A 1×1 projection produces Q, K, V. All queries are retained, while keys and values are gathered only at $\mathcal{I}_K$:

$$Q \in \mathbb{R}^{B\times C\times N}, \tag{23}$$

$$K_s, V_s \in \mathbb{R}^{B\times C\times K}. \tag{24}$$

Attention is

$$A = \mathrm{softmax}\left(\frac{Q^\top K_s}{\sqrt{C}}\right) \in \mathbb{R}^{B\times N\times K}, \tag{25}$$

and context is reconstructed as

$$C_{\text{global}} = V_s A^\top. \tag{26}$$

Thus the main attention matrix scales as $\mathcal{O}(NK)$ instead of $\mathcal{O}(N^2)$ for fixed sparsity ratio.

5.3 Hard Top-K gradient caveat

The historical code discards the selected score values:

```
scores = self.indexer(x)
_, topk_indices = torch.topk(scores, K, dim=1)
```

Only integer indices are consumed downstream. Since the discrete index operation does not provide an ordinary smooth gradient to the score values, the selector should not be assumed to receive a detector-utility learning signal solely from the final attention output. This does not deactivate the global attention path, but it weakens the claim that the indexer itself learns task-optimal saliency.

6 Component III: Engram Associative Memory

6.1 Classification-only intervention

At each FPN scale, the regression branch computes:

$$F_{\text{reg}} \to (\hat{b}, \hat{o}), \tag{27}$$

without memory.

The classification feature

$$F_{\text{cls}} \tag{28}$$

is the only feature modified by Engram.

6.2 Training corruption

During training:

$$F'_{\text{cls}} = (F_{\text{cls}} + \eta) \odot D, \tag{29}$$

where

$$\eta \sim \mathcal{N}(0, 0.05^2), \tag{30}$$

and D is a spatial Bernoulli keep mask with approximately 85% keep probability.

6.3 Latent space and prototypes

Each spatial feature is projected to

$$z_i = W_\ell f_i \in \mathbb{R}^{128}. \tag{31}$$

Each scale owns one prototype per class:

$$P = [p_1, \ldots, p_C]^\top \in \mathbb{R}^{C\times 128}, \tag{32}$$

initialized orthogonally.

Actual retrieval normalizes both query and prototype:

$$\bar{z}_i = \frac{z_i}{\|z_i\|_2}, \qquad \bar{p}_c = \frac{p_c}{\|p_c\|_2}, \tag{33}$$

and uses temperature $T = 50$:

$$a_{ic} = \text{softmax}_c \left(T\, \bar{z}_i^\top \bar{p}_c \right). \tag{34}$$

Retrieved memory is

$$m_i = \sum_c a_{ic} p_c. \tag{35}$$

6.4 Uncertainty-objectness gating

A learned MLP predicts uncertainty

$$u_i = \sigma(h(z_i)). \tag{36}$$

The regression branch provides objectness

$$o_i = \sigma(\hat{o}_i). \tag{37}$$

The memory bank returns

$$m'_i = u_i o_i m_i. \tag{38}$$

After projection back to the classification feature dimension, the head fuses:

$$f_i^{\text{restored}} = (1 - u_i) f_i + u_i W_\ell^\top m'_i. \tag{39}$$

Substituting m'_i shows that the memory term contains approximately

$$u_i^2 o_i, \tag{40}$$

making the historical design more conservative than a single-gate interpretation suggests.

6.5 Memory-anchor supervision

SimOTA produces foreground mask $\mathcal{F}$ and IoU-weighted classification targets y_i^{cls}. The wrapper applies an auxiliary BCE:

$$\mathcal{L}_{\text{mem}} = \text{BCEWithLogits}\left(r_i,\, y_i^{\text{cls}}\right), \qquad i \in \mathcal{F}, \tag{41}$$

where r_i are memory retrieval logits returned by the head.

The wrapper objective is

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{det}} + 0.05\,\mathcal{L}_{\text{mem}}. \tag{42}$$

The small coefficient is important: the detector remains the primary training signal.

7 Detection Loss

The reference version uses standard YOLOX IoU loss:

$$\mathcal{L}_{\text{IoU}} = 1 - \text{IoU}^2. \tag{43}$$

The detector loss is structurally:

$$\mathcal{L}_{\text{det}} = 5\mathcal{L}_{\text{IoU}} + \mathcal{L}_{\text{obj}} + \mathcal{L}_{\text{cls}} + \mathcal{L}_{L1}, \tag{44}$$

where the $L1$ term is active only when configured in the late no-augmentation phase.

No Wasserstein/SWAWIoU term is present in this simpler reference model.

8 Experiment Configuration

Table 1: Core configuration of the supplied simpler TDE experiment.

Setting	Value
Model scale	YOLOX-S
Depth multiplier	0.33
Width multiplier	0.50
Classes	9
Initial TTT LR	0.02
TTT noise std	0.08
Configured maximum epochs	80
Warm-up	10 epochs
No-augmentation phase	15 epochs
Minimum LR ratio	0.05
Weight decay	0.0005
Momentum	0.9
Batch size	16
Gradient accumulation	8
Input/evaluation size	640×640
Test confidence threshold	0.01
NMS threshold	0.65
EMA	enabled
Memory loss weight	0.05
Engram latent dimension	128
Engram temperature	50
Sparse K/V ratio	0.10
Tribrid residual-gate init	0.01

The reported target-condition comparisons are taken at epoch 30, even though the configuration permits longer training.

9 Reported Epoch-30 Results

9.1 Condition-level results

Table 2: TDE versus baseline YOLOX at epoch 30. AP/AR values are percentages; deltas are percentage points.

Cond.	TDE AP	Base AP	ΔAP	TDE AP50	Δ50	TDE AP75	Δ75	TDE AR	ΔAR	ms	Slow
Snow	17.3	16.2	+1.1	34.2	+4.3	16.4	+1.4	27.6	+0.1	20.39	1.80×
Night	12.0	10.7	+1.3	25.7	+2.7	9.2	+0.6	21.8	+1.2	15.89	1.90×
Rain	15.9	14.7	+1.2	27.9	+0.0	17.1	+3.1	27.2	+4.3	15.09	1.94×
Fog	27.4	32.5	-5.1	46.4	-3.9	24.5	-9.0	41.0	-1.4	15.01	1.94×

Across snow, night, and rain:

$$\overline{\mathrm{AP}}_{\mathrm{TDE}} = 15.067\%, \qquad \overline{\mathrm{AP}}_{\mathrm{base}} = 13.867\%, \tag{45}$$

for a gain of approximately

$$+1.200 \text{ percentage points.} \tag{46}$$

Across all four conditions:

$$\overline{\mathrm{AP}}_{\mathrm{TDE}} = 18.150\%, \qquad \overline{\mathrm{AP}}_{\mathrm{base}} = 18.525\%, \tag{47}$$

so fog pulls the four-condition mean to a difference of

$$-0.375 \text{ percentage points.} \tag{48}$$

The scientifically accurate statement is therefore that the model shows *positive transfer on three of four reported adverse conditions*, not that it universally dominates the baseline.

9.2 Scale-specific behavior

Table 3: AP-scale deltas in percentage points, TDE minus baseline.

Condition	$\Delta\mathrm{AP}_S$	$\Delta\mathrm{AP}_M$	$\Delta\mathrm{AP}_L$
Snow	+0.5	-0.6	-3.6
Night	-0.7	+3.1	-0.5
Rain	+0.8	+2.2	-0.6
Fog	-3.0	+1.6	-12.5

Fog is particularly problematic for large objects:

$$\Delta\mathrm{AP}_{75}^{\mathrm{fog}} = -9.0 \text{ points}, \tag{49}$$
$$\Delta\mathrm{AP}_{L}^{\mathrm{fog}} = -12.5 \text{ points}. \tag{50}$$

However, medium-object fog AP increases by +1.6 points. This scale split suggests that the fog failure is not a uniform loss of semantic confidence.

9.3 Selected per-class changes

The supplied evaluator logs show:

- snow bike AP: $0.743 \rightarrow 11.440$ (+10.697 points);
- snow rider AP: $1.443 \rightarrow 5.050$ (+3.607);
- night truck AP: $10.478 \rightarrow 15.172$ (+4.694);
- rain motor AP: $4.545 \rightarrow 12.180$ (+7.635);
- fog bus AP: $58.570 \rightarrow 36.036$ (-22.534);
- fog motor AP: $16.107 \rightarrow 10.540$ (-5.567).

These class-level changes should be treated as diagnostics rather than causal evidence, because class support differs across ACDC subsets.

10 Training Stability Near Epoch 30

The supplied internal evaluator gives:

Table 4: Internal evaluation at epochs 27–30.

Epoch	AP	AP50	AP75	AR100
27	0.249	0.476	0.223	0.367
28	0.249	0.476	0.224	0.369
29	0.250	0.478	0.222	0.371
30	0.251	0.480	0.223	0.371

The model is therefore plateauing smoothly around epoch 30 rather than exhibiting an obvious optimization collapse. This makes the adverse-condition pattern more meaningful: the fog degradation cannot be explained solely by a globally unstable checkpoint.

11 Why the Simpler Architecture Is a Valuable Reference

11.1 Safe authority for the experimental attention path

At initialization:

$$\tanh(g_0) = \tanh(0.01) \approx 0.01. \tag{51}$$

The detector therefore preserves a dominant local path even when the global branch is not yet useful. This is a "safe-to-fail" property.

11.2 Localized adaptation

TTT modifies only the shallow Dark2 normalization-related subset. Deeper backbone stages remain fixed during the one-step adaptation. This limits the capacity of a poor self-supervised objective to rewrite high-level semantics.

11.3 Conservative memory

The Engram module:

- affects classification, not bounding-box regression;
- is anchored to SimOTA foreground targets;
- has auxiliary weight 0.05;
- begins with orthogonal class prototypes;
- uses mild source feature corruption;
- gates memory by uncertainty and objectness.

These choices keep the base detector as the dominant optimization target.

12 Known Implementation Caveats

This section is intentionally explicit because these issues are relevant when interpreting the historical result.

12.1 Mask-ratio naming discrepancy

The "85% mask" uses a 0.85 quantile and masks positions above the threshold, so the masked fraction is approximately 15%, not 85%.

12.2 TTT probability schedule discrepancy

Fields `ramp_end=60` and `prob_end=0.5` are declared, but the active implementation linearly targets 0.7 at the configured maximum epoch.

12.3 Duplicate epoch-state update

The trainer calls the meta-training state synchronization twice in `before_epoch`, once with the current epoch and once with the next epoch. This can shift the TTT probability by one epoch.

12.4 Sparse-indexer gradient limitation

Hard Top-K indices are generated from indexer scores, but score values are discarded. Consequently, the saliency indexer lacks an ordinary differentiable detector-loss pathway through the discrete selection.

12.5 Engram retrieval/supervision geometry mismatch

The actual lookup uses normalized cosine similarity and temperature 50:

$$50\,\bar{z}^\top \bar{p}. \tag{52}$$

The auxiliary logits are instead computed as:

$$z^\top p, \tag{53}$$

without normalization or the same temperature. The auxiliary loss therefore does not exactly supervise the geometry used by retrieval.

12.6 Objectness gradient coupling

Objectness is fed to the memory gate without detachment:

$$o_i = \sigma(\hat{o}_i). \tag{54}$$

Thus memory/classification gradients can in principle propagate through this gate toward the objectness pathway, even though regression features are separately computed.

12.7 Potential batch-order mismatch in memory supervision

This is the most serious latent correctness caveat.

The SimOTA foreground masks are concatenated image-by-image:

$$[\mathrm{img}_0(P3, P4, P5),\ \mathrm{img}_1(P3, P4, P5), \ldots]. \tag{55}$$

The historical memory logits are flattened scale-by-scale:

$$[P3(\text{all images}),\ P4(\text{all images}),\ P5(\text{all images})]. \tag{56}$$

For batch size greater than one, lengths agree but ordering can differ. This can silently pair foreground masks with the wrong memory logits.

A corrected implementation should keep tensors as $[B, N_l, C]$, concatenate scales on N_l, and only then flatten the batch-anchor axes.

12.8 Historical adverse validation and strict ZSDA

The experiment configuration points validation and test annotations at an adverse annotation file. If target-domain validation influences checkpoint selection or hyperparameter design, the procedure is not a strict clean-source-only ZSDA protocol. A strict protocol should use:

$$\text{source train} \rightarrow \text{source validation} \rightarrow \text{freeze} \rightarrow \text{target evaluation}. \tag{57}$$

13 Result Interpretation

13.1 Snow

Snow improves overall AP by +1.1 points and AP50 by +4.3, while AR100 changes by only +0.1. This pattern is compatible with improved classification/confidence/ranking for already detected objects, although component-level ablation is required before attributing the effect specifically to Engram.

Large-object AP decreases by 3.6 points, showing that the gain is not uniformly geometric.

13.2 Night

Night improves AP by +1.3, medium AP by +3.1, and AR100 by +1.2. Small-object AP falls by 0.7. The architecture therefore appears more beneficial to medium-scale nighttime objects than to the smallest instances.

13.3 Rain

Rain improves AP by +1.2, AP75 by +3.1, and AR100 by +4.3. AP50 is unchanged. Large-object recall increases strongly in the supplied log. The result indicates meaningful changes beyond a simple confidence threshold shift.

13.4 Fog

Fog is the critical counterexample. Overall AP falls by 5.1 points, AP75 by 9.0, and large-object AP by 12.5. Medium AP rises by 1.6, so the degradation is scale- and localization-sensitive.

Because the Engram path modifies classification only, a large AP75/large-object failure motivates investigation of shared upstream features, TTT behavior, and neck context rather than treating the memory head as the sole explanation.

14 Causal Claims That the Current Evidence Does Not Support

The supplied full-model comparison does *not* establish that:

- TTT independently improves adverse detection;
- the sparse indexer learns weather-robust saliency;
- Engram alone causes the snow/night/rain gains;
- the three components interact synergistically;
- the model is generally robust across all adverse conditions.

Those claims require controlled ablations.

15 Recommended Reference Ablation Matrix

The historical model should be compared with:

Model	Engram	Sparse/Tribrid	TTT
R0	–	–	–
R1	✓	–	–
R2	–	✓	–
R3	–	–	✓
R4	✓	✓	–
R5	✓	–	✓
R6	–	✓	✓
R7	✓	✓	✓

Table 5: Factorial ablation needed for causal attribution.

16 Reproduction Guidance

16.1 Core run settings

The reproduction should preserve:

- YOLOX-S scale (0.33, 0.50);
- TTT LR 0.02 and noise std 0.08 from the experiment;
- Dark2-only GN adaptation;
- fixed 10% sparse K/V ratio;
- Tribrid gate initialization 0.01;

- Engram latent dimension 128;
- one orthogonal prototype per class per scale;
- temperature 50;
- feature noise 0.05 and spatial dropout 0.15 in classification training;
- memory loss weight 0.05;
- standard IoU loss;
- reported checkpoint epoch 30;
- evaluation size 640×640, confidence 0.01, NMS 0.65.

16.2 Typical YOLOX commands

The exact CLI depends on the checkout. A standard pattern is:

```
python tools/train.py \
  -f exps/default/tde_yolox_s.py \
  -d 1 \
  -b 16 \
  --fp16
```

Condition-specific evaluation typically follows:

```
python tools/eval.py \
  -f exps/example/custom/test_acdc_snow.py \
  -c YOLOX_outputs/<exp>/epoch_30_ckpt.pth \
  -b 1 -d 1 --conf 0.01
```

Paths and flags should be adapted to the actual repository rather than invented from this report.

17 Diagnostics for Future Reproduction

17.1 TTT diagnostics

Record:

- inner reconstruction loss;
- gradient norm of adapted parameters;
- learned per-parameter TTT learning rates;
- $\|F_{\text{after}} - F_{\text{before}}\|_2$;
- same-checkpoint TTT-on versus TTT-off AP.

17.2 Sparse-attention diagnostics

Record:

- $\tanh(g)$ for every Tribrid block;
- Top-K foreground recall;

- clean/perturbed Top-K overlap;
- effective attention entropy;
- indexer gradient norm.

17.3 Engram diagnostics

Record:

- uncertainty distribution on foreground/background;
- objectness gate distribution;
- effective memory contribution;
- prototype cosine-similarity matrix;
- foreground memory-retrieval accuracy;
- per-class memory-anchor loss.

18 Lessons for the Next Architecture

The simpler version suggests that three properties are worth preserving:

1. **Experimental branches should earn authority.** The attention residual begins near zero rather than replacing a large fraction of the local path immediately.
2. **Adaptation should be localized until validated.** Dark2-only norm adaptation limits destructive test-time updates.
3. **Semantic memory should be detector-grounded.** ENGRAM uses actual SimOTA foreground assignments and a small auxiliary coefficient rather than a weather-specific proxy.

Conversely, future versions should correct:

- routing-score supervision / differentiability;
- batch-consistent memory-logit ordering;
- memory retrieval/supervision geometry;
- strict classification/objectness gradient isolation where intended;
- a task-aligned TTT objective;
- source-only checkpoint selection for strict ZSDA.

19 Conclusion

The simpler TDE-YOLOX v2.0 is a technically valuable reference because it produces meaningful positive transfer under snow, night, and rain while retaining a comparatively conservative relationship to the underlying YOLOX-S detector. Its most defensible mechanism is the low-weight, foreground-supervised, classification-only Engram memory. The Tribrid neck is structurally safe because its global branch begins at roughly one-percent amplitude, even though the hard Top-K saliency indexer is not cleanly task-differentiable. The TTT mechanism is shallow and episodic, and its actual masking fraction is much smaller than its comments suggest, but its self-supervised objective is not yet detection-aligned.

The fog regression is an essential part of the evidence: it prevents a universal robustness claim and motivates the next generation of targeted patches. The correct role of this version is therefore as a reproducible reference architecture and ablation anchor from which individual TDE components can be repaired and required to demonstrate independent task utility.

A Implementation Map

File	Responsibility
`yolox/models/darknet.py`	Builds the GN-based Dark2 stage, wraps it in `TTTAdaptiveStage`, and leaves Dark3–Dark5 on standard BN.
`yolox/models/ttt_modules.py`	Implements the projector, functional TTT step, SimAM, Coordinate Attention, MBConv conditioner, sparse attention, Engram bank, and uncertainty estimator.
`yolox/models/tribrid_neck.py`	Implements the local-dense plus sparse-global `C2f_Tribrid` block and its scalar residual gate.
`yolox/models/yolo_pafpn.py`	Replaces the four PAFPN CSP aggregation blocks with Tribrid blocks.
`yolox/models/engram_head.py`	Adds the memory pathway only to the classification branch and returns auxiliary retrieval logits plus SimOTA targets.
`yolox/models/losses.py`	Contains standard YOLOX IoU/GIoU losses; reference run uses $1 - \text{IoU}^2$.
`yolox/models/yolox.py`	Manages TTT probability and the $0.05\,\mathcal{L}_{mem}$ auxiliary loss.
`yolox/core/trainer.py`	Synchronizes epoch state and logs TDE-specific metrics.
`exps/default/tde_yolox_s.py`	Sets model scale, data, TTT parameters, schedule, and model wiring.

B As-Run Versus Corrected Interpretation

Topic	Historical wording/design	Correct technical interpretation
TTT mask	"85% masking"	0.85 quantile threshold with $\geq$ comparison; approximately top 15% masked.
TTT schedule	ramp to 0.5 by epoch 60	active code ramps toward 0.7 at max epoch after warm-up.

Topic	Historical wording/design	Correct technical interpretation
Sparse indexer	learned saliency selector	hard Top-K discards score values; task-gradient pathway to scores is weak/non-smooth.
Engram memory	uncertainty/objectness gated	memory term is effectively gated by uncertainty twice because the bank gates and the outer fusion gates again.
Memory supervision	aligned with SimOTA	target semantics are aligned, but historical flattening can misorder logits and masks for $B > 1$.
ZSDA validation	adverse set available in config	target-informed checkpoint selection would weaken a strict zero-shot claim.

C Reproducibility Checklist

- ☐ YOLOX-S depth 0.33, width 0.50.
- ☐ Nine-class configuration.
- ☐ Dark2 is the only TTT-wrapped backbone stage.
- ☐ Dark3–Dark5 use standard BN blocks.
- ☐ Neck backbone is explicitly overridden with the configured TTT backbone.
- ☐ Four PAFPN aggregation blocks use C2f_Tribrid.
- ☐ Tribrid residual gate initializes to 0.01.
- ☐ Sparse K/V ratio is 0.10.
- ☐ Engram latent dimension is 128.
- ☐ Engram prototypes use orthogonal initialization.
- ☐ Retrieval temperature is 50.
- ☐ Classification feature noise is 0.05 and dropout is 0.15.
- ☐ Standard YOLOX IoU regression is used.
- ☐ Memory auxiliary coefficient is 0.05.
- ☐ Epoch-30 checkpoint is used for the supplied result comparison.
- ☐ Evaluation uses 640×640, confidence 0.01, NMS 0.65.
- ☐ Fog regression is reported rather than omitted from aggregate claims.